K9ESV · McHenry County Emergency Services Volunteers
and McHenry County Emergency Management Agency

FIELDCOMMS

Incident Management System v1.0

COMPLETE USER MANUAL

v1.0

RACES · ARES · Starcom · K9ESV · MCESV / MCEMA

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1 Introduction & System Overview

FieldComms Incident Management System is a self-contained emergency communications server built on a Raspberry Pi 5 for McHenry County RACES, ARES, and Starcom operations. It provides 32 web-based EmComm tools accessible from any smartphone, tablet, or laptop connected to the EMCOMM-NET Wi-Fi access point — no internet, no app installation, and no per-device configuration required. All core features operate fully offline. When internet connectivity is available through InstyConnect cellular or Starlink satellite, live features such as NWS weather alerts, APRS-IS, and HF propagation data activate automatically.

Version 1.0 uses two Raspberry Pi 5 units: the FieldComms application server at 192.168.50.1 running all 32 web tools and 15 background services, and a dedicated AMPRNet gateway Pi at 192.168.50.2 maintaining a permanent WireGuard tunnel into the 44.0.0.0/8 amateur radio IP network. Three ASUS RT-BE58 Go Wi-Fi 7 routers form an AiMesh network: one primary router managing WAN connectivity and DHCP, and two mesh nodes extending EMCOMM-NET coverage to secondary rooms, outdoor staging areas, and upper floors. InstyConnect cellular with T-Mobile and Verizon dual-carrier coverage is the primary WAN source. Starlink satellite provides automatic secondary failover.

1.1 System Architecture

COMPONENT	DESCRIPTION	IP / ACCESS
FieldComms Pi 5 (16 GB)	Application server · Pironman MAX 5 tower · 2× 1 TB NVMe RAID 1 · runs all 32 web pages and 15 Python services	192.168.50.1 / http://192.168.50.1
44Net Gateway Pi 5 (16 GB)	AMPRNet gateway · Argon NEO 5 case · 256 GB M.2 SSD · Pi OS Desktop · WireGuard tunnel to amprgw.ampr.org · routes 44.0.0.0/8	192.168.50.2 / http://192.168.50.2:9000
ASUS RT-BE58 Go (primary)	Wi-Fi 7 access point · DHCP server · dual WAN gateway · AiMesh controller · InstyConnect cellular primary WAN · Starlink automatic failover	192.168.50.254 / http://192.168.50.254
ASUS RT-BE58 Go (2× mesh nodes)	AiMesh coverage extension · same EMCOMM-NET SSID · seamless roaming · wired backhaul via UniFi Switch Ports 11 and 12	Managed by primary router
UniFi Switch Lite 16 PoE	16-port GbE managed switch · 8× PoE · 2× SFP uplink · wired distribution hub for all devices	Layer 2 — no IP
InstyConnect Drum (primary WAN)	Omnidirectional 5G/LTE cellular antenna · modem built into enclosure · T-Mobile + Verizon dual-carrier · single PoE Ethernet to ASUS WAN port	Modem admin: 10.1.1.1
InstyConnect Switchblade (backup antenna)	Directional 4× folding LDAP antenna · swap with Drum cable when signal is poor · aim toward nearest tower using InstyConnect app	Same WAN port as Drum
Starlink dish (automatic failover)	Satellite internet · CGNAT · no inbound connections · connects to ASUS USB WAN port via Ethernet adapter	Dish admin: 192.168.100.1

COMPONENT	DESCRIPTION	IP / ACCESS
Windows Laptop	IC-7300 HF radio · Winlink Express · VARA HF · JS8Call · all connected via single USB cable	192.168.50.3 (recommended)
Pi 500 Workstations (×4)	Browser-based operator stations · Raspberry Pi 500 keyboard computer · Raspberry Pi Monitor 15.6" · USB-C powered	192.168.50.20–23

1.2 Dashboard Modes

The FieldComms dashboard reorganizes its tool cards into three modes selected from the mode bar at the top of the page. The mode changes the layout and highlights the tools most relevant to the current type of operation. All tools remain accessible regardless of mode.

MODE	BEST FOR	HIGHLIGHTED TOOLS
Amateur Radio	ARES/RACES nets, HF operations, Winlink, APRS tracking	Net Control Logger, Callsign Lookup, APRS Map, Propagation, Repeaters, Dead Man Switch
Starcom / Public Safety	Public safety Starcom radio nets, SAR operations, shelter management	Starcom Net Logger, Resource Tracking Map, Resource Board, Facilities Directory
ICS Incident Command	Activated incidents requiring formal ICS documentation and IAP	ICS Platform (all 5 sections), Planning P, ICS-213, ICS-214, Winlink Import

1.3 Offline vs. Online Features

FEATURE	OFFLINE (no WAN)	ONLINE (WAN active)
Net Control Logging	Full — always available	Full — no change
FCC Callsign Lookup	Full — local 800K database	Full — local database used
ICS Platform	Full — all sections	Full — no change
Kiwix Reference Library	Full — stored on Pi	Full — no change
Offline Maps (APRS)	Full — tiles stored on Pi	Full — no change
NWS Weather Alerts	Unavailable	Live — updates every 5 minutes
APRS-IS Feed	RF only via Graywolf/YAAC	Internet APRS-IS active
HF Propagation Data	Last cached data only	Live band conditions
Pat Winlink	RF only (packet/VARA RF)	Internet Winlink gateways
AMPRNet / 44Net	Unavailable (needs WAN)	Live via WireGuard tunnel

2 Getting Started — Connecting to FieldComms http://192.168.50.1/

Getting on FieldComms takes three steps: connect to the Wi-Fi, open the dashboard, and identify yourself. The whole process takes under a minute.

Step 1 — Connect to the EMCOMM-NET Wi-Fi

On your phone, tablet, or laptop, open Wi-Fi settings and connect to:

SETTING	VALUE
Network Name (SSID)	EMCOMM-NET
Password	Provided by your net manager (default: fieldcomms2026)
Security	WPA2-PSK
Wi-Fi Channel	6 (2.4 GHz)
Your device receives IP	192.168.50.100 – 192.168.50.200 (automatic)

The Wi-Fi password is the only access control on FieldComms. Anyone with the password can use every tool — there is no separate login screen. Keep the password posted at the EOC for arriving operators and treat it as an operational credential.

Step 2 — Open the Dashboard

1. Open any web browser (Chrome, Safari, Firefox, Edge — all work).
2. Type **http://192.168.50.1** in the address bar and press Enter.
3. The FieldComms dashboard loads. You do not need to install anything.

Step 3 — Identify Yourself

On first visit, a prompt appears asking for your identity. Enter your **callsign, Radio ID, or name** and optionally your ICS position. This information is saved in your browser and remembered between sessions.

You do not need to re-enter your identity every time you visit. Your browser remembers it until you clear browser storage or tap the identity badge to change it.

3 The Main Dashboard http://192.168.50.1/

The dashboard is your central hub for every activation. The moment you open http://192.168.50.1 in any browser on EMCOMM-NET, you see the full picture: live NWS weather alerts with severity color coding, the real-time APRS station table from Graywolf and YAAC, system health indicators, and the Dead Man Switch state for any active net. At the very top is the MCESV/MCEMA organization header and your operator identity badge. The three-button mode switcher below it reorganizes the tool cards to match the type of operation you are running.

Three Dashboard Modes

Directly below the header is a three-button mode switcher. It reorganizes the tool cards to match what you are doing. Your choice is saved on your device and remembered between sessions.

MODE	BUTTON	WHAT IT EMPHASIZES	WHEN TO USE
■ Amateur Radio	■	Net Control, APRS, Winlink, Callsign Lookup, Resource Board, Roster, DMS, Pre-Flight, NTS/ICS forms, Propagation, Repeaters, Kiwix, Cheat Sheets	ARES/RACES nets, NTS traffic, weekly nets, exercises
■ Starcom / Public Safety	■	Starcom Net Logger, Weather Net, SAR Net, Observer Mode, Resource Tracking Map, Resource Board, Member Roster, Facilities, ICS Platform, ICS-213, ICS-214	Starcom activations, public safety nets, weather spotters, SAR operations
■ ICS / Incident Command	■	ICS Platform (all five sections + Planning P), Tactical Map, Resource Map, Resource Board, Roster, Facilities, ICS forms, Winlink Import, Repeaters, Cheat Sheets	Formal ICS activations, EOC operations, multi-agency incidents

Dashboard Elements

ELEMENT	DESCRIPTION
Hero bar	Station callsign badge, system name, live UTC clock and local 24-hour clock updated every second
Mode switcher	Three mode buttons — ■ Amateur Radio / ■ Starcom / ■ ICS — click to reorganize tool cards
NWS Weather Alerts	Live National Weather Service alerts color-coded by severity. Click any alert to expand full details, onset time, expiry countdown, and protective action instructions
Operation cards	Quick-launch tiles for every tool in the current mode — click to open
APRS station table	Live APRS stations heard by Graywolf, with callsign, distance, course/speed, comment, and last heard time. Updates every 30 seconds
Status sidebar	CPU/memory/disk/temperature, all service status dots (green=running, red=stopped), GPS status, Dead Man's Switch state

■ The URL parameter **?mode=starcom** loads the dashboard directly in Starcom mode — useful for bookmarking on Starcom-dedicated devices. Similarly, **?mode=ics** opens directly in ICS mode for dedicated incident command tablets.

2.3 Connectivity Status Cards

Two live connectivity status cards are visible at the bottom of all three modes:

CARD	SHOWS	LINKS TO
WAN Status	Active WAN source — InstyConnect cellular, Starlink satellite, or offline. Card turns green for cellular, blue for Starlink failover, red for offline. Updates every 30 seconds.	wan-status.html — full WAN dashboard with signal strength, carrier, latency, connectivity tests, and WAN event log
AMPRNet Gateway	WireGuard tunnel state — UP with 44.x.x.x AMPRNet address shown, or TUNNEL DOWN in red. Updates from the gateway Pi every 30 seconds.	amprgate.html — full gateway dashboard with tunnel controls, traffic stats, and routing table

2.4 Health Monitor Sidebar

The right panel on the dashboard shows live system health: CPU temperature, memory and disk usage, GPS fix status, internet connectivity, and a row of colored service status dots — one per background service. A green dot means the service is running normally. A red dot means it has stopped or failed. Note the service name from the dot tooltip, then SSH to the Pi and run sudo systemctl restart [service-name] to restart it. The health panel refreshes every 15 seconds automatically.

4 Member Roster http://192.168.50.1/roster.html

The Member Roster is the authoritative directory for all MCESV/MCEMA personnel. It tracks every member's identifiers, contact information, certifications, equipment, and deployment activations. It supports both amateur radio operators and non-ham members.

Member Identifiers

Every member is tracked by up to three identifiers:

IDENTIFIER	WHO HAS IT	EXAMPLE
ESV Member ID	All ESV members	ESV-042
Starcom Radio ID	All ESV members	1042
Amateur Callsign	Licensed hams only (blank for non-hams)	K9ESV

Non-ham members are full members of the roster. Leave the Callsign field blank for them — the system uses their Radio ID or Member ID as their primary identifier automatically.

The Roster Tabs

TAB	CONTENTS
Directory	Full searchable member list with all identifiers, contact info, and role badges
Certifications	Check off ICS-100/200/300/400/700/800, EmComm I/II, CERT, First Aid, FEMA IS courses
Equipment	Check off capabilities: HF/VHF/UHF radio, Winlink, JS8Call, APRS, generator, antenna, laptop, go-kit
Activations	Check-in log for the current activation. Click + Check In to mark a member as activated. Walk-in check-ins are added here
Import/Export	Export roster to CSV for backup or sharing. Import from a CSV to bulk-load members

Role Definitions

ROLE	DESCRIPTION
Net Control (NCS)	Primary net controller — manages check-ins and traffic flow
Operator	Standard ARES/RACES field operator
Liaison	Agency liaison — interface between operators and the served agency
Emergency Coordinator (EC)	Oversees ARES/RACES operations for the county

Importing & Exporting (CSV)

1. To export: click **Export CSV**. A file downloads with all members and all columns.

2. To import: prepare a CSV with these column headers, then click **Import CSV** and choose the file.

```
member_id, callsign, radio_id, first_name, last_name, role, phone, email, grid, license_class
```

■ The net manager can print wallet-sized access cards for every member, generated from the roster. Each card shows the Wi-Fi network and password, the dashboard URL, and the member's identifiers. Run: `python3 /opt/fieldcomms/scripts/gen_operator_cards.py --ssid EMCOMM-NET --password YOUR-PASSWORD`. Output is a print-ready PDF, 10 cards per Avery 5371 sheet.

3.2 CSV Import Column Reference

The roster accepts CSV files exported from any spreadsheet application. The following column headers are recognized — extra columns are ignored:

COLUMN HEADER	REQUIRED	DESCRIPTION
callsign	Yes	FCC amateur callsign — auto-filled in net logs via FCC database lookup
radio_id	Yes	Starcom Radio ID number — used in Starcom net logger check-ins
first_name	Yes	Operator first name
last_name	Yes	Operator last name
role	No	Role or position — Net Control, Field Unit, EOC Staff, etc.
phone	No	Contact number for off-air coordination
email	No	Email for Winlink or off-air messaging
certifications	No	Comma-separated — NIMS-100, NIMS-700, SKYWARN, etc.

■ Export CSV from Excel: File → Save As → CSV UTF-8 (Comma delimited). Export from Google Sheets: File → Download → Comma Separated Values. UTF-8 encoding is required — other encodings may cause character errors in names with accented characters.

5 Operator Identity System

Every FieldComms page tracks who is operating at each device so your check-ins, log entries, and form signatures are attributed correctly. There is no password — your identity is stored in your device's browser and remembered between sessions. The system handles all four kinds of people who show up at an activation.

The Four Identity Types

TYPE	IDENTIFIERS USED	BADGE
ESV Member + Ham	Member ID + Callsign + Starcom Radio ID	■ K9ESV
ESV Member, Non-Ham	Member ID + Starcom Radio ID (no callsign)	■ ESV-042
Visitor / Mutual Aid, Ham	Callsign + Agency name	■ W9XYZ
Visitor / Mutual Aid, Non-Ham	Name + Agency (+ optional Radio ID)	■ VISITOR

The system always displays the most useful identifier for radio work. The priority is: callsign first, then Starcom Radio ID, then member ID, then name.

Setting Your Identity

1. On first visit, a prompt appears automatically. Enter your callsign, Radio ID, or name, and optionally your ICS position.
2. Click **Set Identity**. Your badge appears in the header.
3. To change your identity later: click your callsign/name badge in the header and update the fields.

> ■ Observer Mode (http://192.168.50.1/observer.html) intentionally does NOT prompt for identity — it is read-only and requires no identification.

6 Amateur Net Control Logger http://192.168.50.1/netcontrol.html

The Amateur Net Control Logger is the primary tool for running ARES/RACES amateur radio nets. When a station checks in, you type their callsign and press Enter — FieldComms looks up the operator in the local FCC database and fills their name and license class automatically. The check-in is timestamped in UTC and saved to the server immediately, making it visible to all connected devices including served-agency staff watching in Observer Mode.

The logger supports multiple nets running simultaneously, each on its own tab, each with an independent check-in log, traffic log, and ICS-309 export. A weekly ARES net, a RACES emergency activation, and an NTS traffic session can all be logged at the same time with one operator on the keyboard.

Starting a Net

1. Click **+ New Net** in the net tabs bar at the top.
2. Enter a **Net Name** — for example "McHenry County ARES Thursday Evening Net".
3. Enter the **Frequency** — for example "147.015 MHz / 107.2 Hz".
4. Enter **Net Control** — your callsign (auto-filled from your identity if set).
5. Click **Activate Net**. The net goes live and appears as a tab.

The Net Selector Tabs

Each active net appears as a tab in the NET TABS bar at the top of the page. The active net is highlighted. Tap any tab to switch to that net's log. Each net has its own independent check-in log, traffic log, and export.

Logging a Check-In

1. Type a callsign in the **Callsign** field and press **Enter**. The FCC database fills in the operator's name and license class automatically.
2. Add **Location** and **Remarks** if needed.
3. Click **+ Check In** or press Enter. The entry appears with a UTC timestamp.
4. To add a non-licensed or non-US operator: type their name or identifier directly.

> ■ The **+ Roster** button only appears for stations not already in the roster, so you will not create duplicates. Click it to add any new station to the Member Roster directly from the net log.

Logging Traffic

1. Switch to the **Traffic** tab.
2. Enter the From callsign, To callsign or address, traffic type, and a note.
3. Click **Log Traffic**. The traffic item is timestamped and recorded with the net.

The Roster Chips (Quick Check-In)

The Roster tab shows quick-tap chips for known stations. Tap a chip to instantly log that station's check-in without retyping the callsign — useful for regulars on a weekly net.

Sharing & Exporting

BUTTON	WHAT IT DOES
■ Observer Link	Copies a read-only URL for the currently selected net. Share it with served-agency staff or EOC viewers — see Chapter 8.
ICS-309	Exports the selected net's log as a formatted ICS-309 Communications Log, ready to print.
Export JSON	Downloads the full net data as a JSON backup file.
Close Net	Marks the net closed and archives it. The log is preserved and viewable.

> Precedence levels — ROUTINE / WELFARE / PRIORITY / EMERGENCY — can be set per check-in or traffic entry. Use EMERGENCY only for life-safety traffic. PRIORITY for time-sensitive but non-emergency messages.

7 Starcom Net Logger http://192.168.50.1/starcom.html

The Starcom Net Logger has its own dedicated dashboard mode — separate from the Amateur Radio section. Select ■ **Starcom / Public Safety** from the mode bar on the main dashboard to access it. The module is purpose-built for public safety Starcom radio net logging and handles the identifiers, terminology, and workflow that public safety operators use day to day. Units are identified by Radio ID and unit number rather than amateur callsigns. A dispatch center field tracks which agency is managing traffic on each channel. Net names use the sc- prefix by convention so they appear clearly labeled in exports and the Dead Man Switch monitor.

Running Multiple Simultaneous Nets

The Starcom Net Logger supports multiple simultaneous nets — identical to the Amateur Net Control Logger. This is particularly useful during large activations where you may need to run a **Starcom General Net**, a **Weather Net**, and a **SAR Net** all at the same time, each with its own independent check-in log and traffic log.

1. Click **+ New Net** to open the first net.
2. Click **+ New Net** again for each additional net — Weather Net, SAR Net, etc.
3. Each open net appears as a clickable badge in the **active nets panel** on the left sidebar.
4. Click any net badge to switch to that net. The check-in log and traffic log update to show that net's data.
5. Nets run independently — logging a check-in on one net does not affect the others.
6. Close each net individually when it is no longer needed.

> ■ Typical multi-activation scenario: Starcom General Net (command channel) + Weather Net (storm spotters) + SAR Net (field teams) all open simultaneously. Net Control switches between them as radio traffic comes in on each channel.

Opening a Starcom Net

1. Click **+ New Net**.
2. Enter the **Net Name**.
3. Choose the **Net Type**: Starcom General / Weather Net / SAR Net / Observer Net.
4. Enter the **Talkgroup** and **Channel/Frequency**.
5. Enter the **Dispatch Center** (e.g. MCECC, IDOT). It appears in the net header and stays set for the whole net session.
6. Click **Activate Net**.

Logging a Unit Check-In

1. Enter the **Radio ID / Unit #** (e.g. 1234).
2. Enter the **Unit Name / Callsign** (e.g. "MCHENRY CO ARES" or a callsign).
3. Choose the status: Check-In, Traffic, Priority Traffic, Emergency, Dispatch, Check-Out, En Route, On Scene, Available, or Out of Service.
4. Select the talkgroup type and enter the channel/frequency.
5. Add location and remarks, then click **LOG ENTRY**. The time is stamped automatically.

Exporting & Sharing

The Starcom logger shares the same Observer Link, ICS-309 export, and JSON backup tools as the amateur logger (see Chapter 6). Exports carry a Starcom header and keep the sc- prefix.

FIELD	DESCRIPTION
Radio ID	Starcom unit number (e.g. 2301, 4710). Required for every check-in.
Unit Name	Agency and unit description (e.g. "McHenry SO — Unit 12")
Dispatch Ctr	Dispatching agency or center (e.g. MCECC, IDOT)
Net Type	Starcom General / Weather Net / SAR Net / Observer Net
Talkgroup	Radio talkgroup identifier for the net channel

8 Observer Mode — Read-Only Net View http://192.168.50.1/observer.html

Observer Mode provides a read-only, auto-refreshing view of any active net. It is designed for served-agency personnel, EOC staff, emergency managers, and section leaders who need to monitor radio traffic without touching the net control interface. No login, no identity prompt, no setup required.

What Observer Mode Shows

ELEMENT	DESCRIPTION
Net name & status	Active net name, type (Amateur/Starcom), and open/closed indicator
Live check-in log	All stations or units checked in, with timestamp, location, and remarks
Traffic log	Formal message traffic passed during the net
Station count	Live count of checked-in stations at the top of the page
Auto-refresh	Page refreshes automatically every 15 seconds — no manual reload needed

How to Open Observer Mode

1. On the Net Control Logger or Starcom Logger page, click ■ **Observer Link** to copy the URL to your clipboard.
2. Send the link by email, Winlink, JS8Call, or read it over the air to any operator who needs to monitor.
3. The recipient opens the link in any browser on EMCOMM-NET — phones, tablets, and laptops all work.
4. Observer Mode shows the net immediately with no identity prompt and no login.
5. The net name is encoded in the URL so the link can be bookmarked for recurring nets.

■ Observer Mode is completely passive — observers cannot add check-ins, log traffic, or close the net. If an observer needs to interact, they must switch to the Net Control Logger or Starcom Logger and identify themselves first.

9 FCC Callsign Lookup http://192.168.50.1/callsign.html

The Callsign Lookup page provides instant offline access to the full FCC Amateur Radio ULS database — approximately 800,000 active licensees. No internet connection is required. Lookups are answered in milliseconds from the local SQLite database on the Pi.

Quick Callsign Lookup

1. Type a callsign (e.g. K9ESV) in the search box at the top of the page.
2. Press **Enter** or click ■ **Look Up**.
3. The result shows: full name, license class, expiration date, mailing address, grid square, and FRN.
4. Click **+ Add to Roster** to add the licensee directly to the Member Roster.
5. Recent lookups are saved automatically and shown below the search box for quick re-access.

Advanced Search

1. Click the **Advanced Search** tab.
2. Enter any combination of: First Name, Last Name, City, State, License Class, or Grid Square.
3. Click **Search**. Results show up to 100 matching licensees.
4. Click any result row to see the full record.
5. Click **+ Roster** on any result to add them to the Member Roster.

■ The FCC database is refreshed automatically every Sunday at 03:00 via the **fcc-refresh.timer** systemd timer (requires internet connection). The database status — record count and last update date — is shown at the bottom of the page.

10 Tactical APRS Map http://192.168.50.1/tactical.html

The Tactical APRS Map is a full-screen interactive map that combines live APRS station tracking with manual tactical overlays and offline map tiles. It pulls station data simultaneously from Graywolf (port 8080) and YAAC (port 8082), merges the feeds, deduplicates by callsign, and keeps every connected browser updated via WebSocket. The map works completely offline once tiles have been downloaded — no internet connection is needed to see stations or draw overlays.

The default map layer is USGS Imagery+Topo Hybrid — public-domain satellite imagery with roads, contours, and place names overlaid. Additional tile sets (OpenStreetMap, Esri World Imagery) are available in the layer selector. McHenry County map tiles are included with the FieldComms installation. Additional county and state tile sets are downloaded using the download_tiles.sh script.

Map Layout

ELEMENT	DESCRIPTION
Left sidebar	Station list — all APRS stations received, with callsign, comment, distance, and last heard time
Map area	Interactive Leaflet map with APRS symbols, custom overlays, and drawing tools
Top toolbar	Layer toggles, tile selector, auto-refresh interval, range ring, and drawing tools
Station labels	Callsign labels on the map — toggleable in settings
Status bar	Station count, GPS status, last refresh time

APRS Sources — Graywolf and YAAC

SOURCE	PORT	DESCRIPTION
Graywolf	8080	Primary APRS client — receives RF APRS via connected TNC or soundcard interface
YAAC	8082	Secondary APRS client — backup receive and alternate TNC support

Station Tracking

1. Click any station in the left sidebar to center the map on that station.
2. Click ■ **Track** to lock the map view on a specific station as it moves.
3. Use the **Station age threshold** setting to hide stations older than N minutes.
4. The search box filters the station list by callsign or comment text.

Drawing Tactical Overlays

1. Click a drawing tool in the toolbar: **Marker**, **Polygon**, **Circle**, or **Route**.
2. Click on the map to place the shape. For polygons and routes, click multiple points and double-click to finish.
3. Click a placed marker or shape to add a label or delete it.
4. Overlays are saved to the server automatically and persist across browser sessions.

5. Click **KML Export** to download the current overlay set as a KML file.

GPS Position

If a USB GPS receiver is connected to the Pi, your current position appears on the map as a blue dot and is used to calculate distances in the station list. The GPS status indicator in the status bar shows the fix quality.

Range Rings

Enter a radius in km in the **Range ring** field and click **Draw Ring** to place a circle on the map centered on your GPS position or the station default coordinates. Useful for visualizing coverage areas.

9.4 Map Layers

LAYER	DESCRIPTION	AVAILABILITY
USGS Topo + Imagery (default)	USGS satellite imagery with roads, contours, and place names overlaid. Best for field operations and SAR.	Offline — tiles stored on Pi
OpenStreetMap	Detailed street map with building outlines, trails, and POIs. Good for urban EOC operations.	Offline — tiles stored on Pi
Esri World Imagery	High-resolution satellite imagery without overlays. Best for aerial reconnaissance and damage assessment.	Online only — requires WAN
NOAA NWS Radar (overlay)	Live NOAA radar precipitation overlay on any base layer. Updates every 5 minutes when WAN is available.	Online only — requires WAN

■ Offline map tiles for McHenry County and surrounding counties are pre-loaded during installation. To download tiles for additional counties or states, run: sudo bash /opt/fieldcomms/scripts/download_tiles.sh --region [region-name]. Each county tile set uses approximately 200-500 MB of storage.

11 Starcom Resource Tracking Map http://192.168.50.1/resmap.html

The Resource Tracking Map is a purpose-built situational awareness tool for Starcom public safety net operations. It shows unit positions on an offline map with status color coding, supports manual placement and drag-and-drop repositioning, and synchronizes with the Member Roster for unit information. Access it from the Starcom / Public Safety dashboard mode.

Placing a Unit on the Map

1. Click **+ Add Unit** in the left panel.
2. Enter the **Radio ID**, **Unit Name**, and **Type** (Law Enforcement, Fire, EMS, Search & Rescue, Communications, EOC, Shelter, Other).
3. Enter a **Latitude** and **Longitude** directly, or click ■ **Pick on Map** to click the map and set the position.
4. Set the **Status** (Available, Deployed, Staging, Out of Service).
5. Click **Save**. The unit appears on the map with its status color.

Status Colors

STATUS	COLOR	MEANING
Available	Green	Unit is on scene and available for assignment
Deployed	Blue	Unit is actively assigned and in the field
Staging	Amber	Unit is staging or en route
Out of Service	Red	Unit is unavailable

Moving Units

1. Click and drag a unit marker on the map to a new position.
2. The unit's coordinates update automatically when you release.
3. Alternatively, click the unit marker, click **Edit**, and update the coordinates manually.

Drawing Zones

1. Click **Draw Zone** to enter zone-drawing mode.
2. Click multiple points on the map to define the zone boundary.
3. Double-click to finish. Enter a zone name and color when prompted.
4. Zones persist across browser sessions and appear on the map for all connected devices.
5. Click **Clear Zones** to remove all zones.

12 Resource Board http://192.168.50.1/resources.html

The Resource Board tracks all personnel, vehicles, equipment, and facilities assigned to an activation. Each resource is displayed as a card showing its current status. A stats strip at the top of the page shows live counts by status across all resource types.

The Five Status States

STATUS	COLOR	MEANING
Available	Green	On scene and available for assignment
Assigned	Amber	Actively tasked with a specific assignment
Staging	Blue	En route or at the staging area, not yet deployed
Out of Service	Red	Unavailable — mechanical, medical, or administrative hold
Demobilized	Gray	Released from the incident and departed

Adding a Resource

1. Click **+ Add Resource**.
2. Enter the **Name/Description** (e.g. "IC-7300 HF Radio", "Unit 12 — Medic", "Generator — 5kW Honda").
3. Select the **Type**: Personnel, Vehicle, Radio, Generator, Antenna, Computer/Tablet, Medical, Shelter/Tent, Repeater, or Other.
4. Set the **Status** and optionally fill in Owner/Callsign, Quantity, Location, Assigned To, Contact, and Notes.
5. Click **Save Resource**. The card appears immediately.

Changing a Resource Status

1. Click the colored **status badge** on any resource card.
2. Each click cycles through the five states in order: Available → Assigned → Staging → Out of Service → Demobilized → Available.
3. The stats strip at the top updates instantly.

Filtering the Board

1. Use the **Type** filter dropdown to show only Personnel, Vehicles, Equipment, etc.
2. Use the **Status** filter to show only Available or only Assigned resources.
3. Filters can be combined — e.g. show all Assigned Personnel only.
4. The stats strip always reflects all resources regardless of the active filter.

13 Dead Man's Switch — Net Inactivity Monitor

http://192.168.50.1/deadmans.html

The Dead Man's Switch (DMS) is a net inactivity monitor. When armed for a specific net, it watches the last-activity timestamp on that net. If no check-in, no traffic entry, and no manual reset occurs within the configured threshold period, the DMS fires: an audible alarm sounds on every browser that has the DMS page open, the page background turns red, and a countdown flashes until the situation is resolved. This gives net control operators an automatic safety net against a silent radio, a dropped connection, or an unattended console during a long activation.

The DMS can be armed independently for each active net running in FieldComms. A major activation running a Starcom General Net, a Weather Net, and a SAR Net simultaneously can have each net on its own DMS with its own threshold. The DMS page runs in the browser tab where it was opened — keep it visible on a dedicated monitor or the net control operator display.

12.1 How to Arm the DMS

1. Open http://192.168.50.1/deadmans.html in a browser on any EMCOMM-NET device.
2. Select the net you want to monitor from the dropdown. All active nets in FieldComms appear here automatically.
3. Set the threshold — how many minutes of inactivity before the alarm fires. Typical settings: 10 minutes for a busy net, 15 minutes for a weather or traffic net, 5 minutes for a SAR net where frequent check-ins are expected.
4. Click ARM. The countdown timer starts from the last recorded activity on that net.
5. Leave the DMS tab open and visible. The alarm fires in this tab and browser only. For maximum coverage, open the DMS on a dedicated monitor at the net control position.

12.2 When the DMS Fires

When the inactivity threshold is exceeded the DMS fires immediately. The browser tab turns red, a loud alarm tone plays, and the countdown shows how long ago the last activity was recorded. Any of the following actions resets the DMS and stops the alarm:

RESET ACTION	HOW	NOTES
Station check-in	Any operator checks into the net using the Net Control Logger	Most common reset during a normal net — just running the net keeps the DMS satisfied
Traffic entry	Any operator adds a traffic item to the net traffic log	Logging a message or status update counts as activity
Manual reset	Click the RESET button on the DMS page	Use during planned quiet periods — briefings, breaks, or scheduled silence
Disarm	Click DISARM on the DMS page	Disarm when the net is formally closed or monitoring is no longer needed

12.3 Multiple Nets

Each net in FieldComms can have its own independent DMS instance. Open http://192.168.50.1/deadmans.html in separate browser tabs, one per net, and set different thresholds for each. For example: the Starcom General Net at 10 minutes, the RACES HF net at 20 minutes during an extended overnight activation, and the SAR Coordination Net at 5 minutes. Each tab monitors and alarms independently.

> The DMS runs in the browser tab — it does not send alerts to other devices. If the operator closes the tab or the device goes to sleep, the DMS stops. For overnight or extended activations, use a dedicated device on AC power with screen-saver and sleep disabled for the DMS monitor.

14 Pre-Flight Deployment Checklist http://192.168.50.1/preflight.html

The Pre-Flight Checklist is a structured GO / CAUTION / NO-GO deployment readiness assessment. Before any activation, run through the checklist to verify that all critical systems are operational.

Checklist Sections

SECTION	ITEMS COVERED
Power Systems	Shore/generator power, battery backup/UPS, solar panels, fuel supply, power distribution
Communications Equipment	HF radio, VHF/UHF radio, go-kit radios, HF antenna, VHF antenna, repeater access, Winlink, JS8Call
Computing & Software	Pi server boot, FCC database, nginx, API server, health monitor, paper backup forms
Networking	ASUS router, EMCOMM-NET Wi-Fi, DHCP, field device connectivity, UniFi switch
Field Supplies	Operator cards, printed forms, batteries, cables, tools, maps, personal PPE
Coordination	Net frequency confirmed, EOC contact established, served agency notified, ICS structure assigned

Deployment Verdicts

VERDICT	MEANING
■ GO	All required items passed — system is ready for deployment
■ CAUTION	Some optional items failed — review before deploying
■ NO-GO	One or more required items failed — do not deploy until resolved

Exporting the Report

1. Click **Export Report** to generate a printable PDF summary of the checklist.
2. The report includes the verdict, timestamp, and notes for all items.
3. File the report with the incident documentation.

■ Run the Pre-Flight Checklist before every activation — even for exercises. It takes less than five minutes and catches configuration issues before they become problems during a real emergency.

15 ICS Platform — Overview http://192.168.50.1/ics/

The ICS Platform is the full incident command and documentation system built into FieldComms. It covers all five standard ICS sections — Command, Operations, Planning, Logistics, and Finance/Admin — plus an interactive Planning P cycle guide. All five sections share a single active incident record, so changes made in Operations are immediately visible in Planning, and the Logistics comms plan feeds directly into printed IAP packages.

The ICS Platform is accessible by selecting the ICS mode from the dashboard mode bar, or directly at http://192.168.50.1/ics/. All data is stored on the Pi and synchronized in real time across every device on EMCOMM-NET. Section chiefs at different tables in the EOC all see the same incident state simultaneously without refreshing their browsers.

Creating a New Incident

1. From the main dashboard (ICS mode), click ■ **New Incident**.
2. Enter the Incident Name (required), Type, Location, Jurisdiction, Incident Commander, Op Period Duration, and an optional Situation Summary.
3. Click **Create Incident**. The incident becomes the active incident for all ICS platform pages.

ICS Navigation Tabs

TAB	SECTION	PRIMARY USE
■ Command	Command Section	Incident objectives, safety, weather, situation
■ Operations	Operations Section	T-card board, resource assignments
■ Planning	Planning Section	IAP documents, resource status table, period tracking
■ Logistics	Logistics Section	Comms plan, supply tracking, meals
■ Finance	Finance / Admin Section	Cost accounting, time tracking, procurement
■ Planning P	Planning Cycle Guide	Step-by-step ICS planning cycle with forms and attendees

Supported Incident Types

TYPE	TYPICAL USE
Natural Disaster	Severe weather, flooding, earthquake, tornado response
HazMat	Chemical spill, gas leak, radiological incident
Search & Rescue	Lost person, overdue hiker, water rescue
Mass Casualty Incident	Multi-patient medical emergency, transportation accident
Weather Net	Starcom/SKYWARN weather spotter activation

TYPE	TYPICAL USE
Public Health	Disease outbreak, mass vaccination, shelter-in-place
Other	Any incident not covered by the above types

14.3 Multi-User Real-Time Collaboration

Every device on EMCOMM-NET connected to the ICS Platform sees the same incident state simultaneously. Changes made by the Operations Section Chief on one tablet appear immediately on the Planning Section Chief tablet and the Logistics Section Chief laptop without anyone pressing a refresh button. This real-time synchronization uses WebSocket connections maintained by the ics-platform background service on the Pi.

The recommended workflow for a fully-staffed EOC activation is:

POSITION	DEVICE	ICS SECTION
Incident Commander	Tablet on EMCOMM-NET	ics/command.html — objectives, safety, public info
Operations Section Chief	Laptop on EMCOMM-NET	ics/operations.html — T-card board, resource assignments
Planning Section Chief	Laptop on EMCOMM-NET	ics/planning.html — IAP tracker, situation status
Logistics Section Chief	Tablet on EMCOMM-NET	ics/logistics.html — comms plan, supply requests
Finance/Admin	Laptop on EMCOMM-NET	ics/finance.html — cost tracking, time log
Net Control Operator	Pi 500 workstation	netcontrol.html + starcom.html — radio net logging

16 ICS Command Section http://192.168.50.1/ics/command.html

The Command Section captures the strategic-level information for the incident: incident objectives, safety hazards, weather summary, current situation, accomplishments, and planned actions. This content forms the core of the Incident Action Plan (IAP).

Incident Objectives (ICS-202)

1. To add a pre-defined objective: click the **Select a common objective** dropdown. Choose from 100 common ICS objectives organized in 13 groups.
2. The selected objective populates the text field below. Edit it if needed — fill in any bracketed placeholders such as [frequency], [location], or [time].
3. If no edits are needed, click **+ Add** or press Enter.
4. To add a completely custom objective: type it directly in the text field and click **+ Add**.
5. Objectives are listed above the input field. Click the ✕ next to any objective to delete it.

> ■ The 100-item objectives dropdown is organized into 13 groups: Life Safety, Communications, Resource Management, Incident Command, Search & Rescue, Shelter & Mass Care, Weather & Natural Disaster, HazMat, Mass Casualty, Public Information, Finance & Administration, Demobilization, and MCESV/MCEMA Specific.

Situation Report Fields

FIELD	WHAT TO ENTER
Safety Message	Safety message for this operational period (ICS-208)
Current Situation	Brief description of current incident status and key facts
Accomplishments	What has been achieved so far this operational period
Planned Actions	What operations are planned for the next operational period
Weather Summary	Current and forecast weather affecting operations

17 ICS Operations Section http://192.168.50.1/ics/operations.html

The Operations Section manages the tactical resources assigned to the incident. It uses a T-card style status board where resource cards can be dragged between status columns, mirroring a traditional ICS T-card board.

The T-Card Board

COLUMN	MEANING
Available	Resource is checked in and available for assignment
Assigned	Resource has been given a tactical assignment
Out of Service	Resource is unavailable — mechanical, medical, or other
Staging	Resource is at the staging area awaiting assignment
Released	Resource has been demobilized from the incident

Adding a Resource to the Board

1. Click **+ Add Resource** or **+ Assignment**.
2. Enter the resource name, type, callsign/Radio ID, and initial assignment.
3. Click **Save**. The resource appears in the Available column.

Moving Resources Between Columns

1. Drag a resource card from one column to another to update its status.
2. Alternatively, click the resource card to open its detail view, then change the status in the dropdown.
3. All status changes are saved to the server immediately.

Assignment Details

1. Click any resource card to open the detail panel.
2. Enter or update the **Assignment** (Division/Group), **Supervisor**, **Work Location**, and notes.
3. This information feeds the ICS-204 Assignment List.
4. Click ■ **Save** to commit the changes.

18 ICS Planning Section http://192.168.50.1/ics/planning.html

The Planning Section manages the Incident Action Plan documents, resource status summary table, and operational period tracking. It is where the PSC assembles all the pieces of the IAP before each operational period briefing.

IAP Tracker Tab

1. Click the **IAP Tracker** tab.
2. Each required form (ICS-202 through ICS-208) is listed with a completion checkbox.
3. Check off each form as it is completed and added to the IAP package.
4. The tracker shows the percentage of IAP completion at the top.

Resource Status Summary

1. Click the **Resource Status** tab.
2. The table shows all resources by type with current status counts.
3. Click **+ Add Row** to manually add a resource category.
4. Counts update from the Operations Section T-card board automatically.

Incident Documents

1. Click the **Documents** tab to manage IAP-associated documents.
2. Click **+ Add Document** to link an uploaded reference document to this incident.
3. Enter the document title, form number, and notes.
4. Documents are linked to the incident for archival purposes.

> The ICS-309 Communications Log (http://192.168.50.1/ics309.html) can be linked to any active incident from its own page. See Chapter 22 for details on the ICS-309 form.

19 ICS Logistics Section http://192.168.50.1/ics/logistics.html

Logistics provides all facilities, services, and materials that support the incident. It owns the communications plan and tracks supplies, food, medical, facilities, and check-in.

TAB	HOW TO USE IT
Comms Plan (ICS-205)	Build the radio communications plan. Each row is a channel: function, channel name, frequency, CTCSS tone, mode, remarks. Click **Add Comms Row** to add a channel. Pre-filled with McHenry County Starcom channels.
Supply	Log supply requests: item, category, quantity needed, quantity on hand, priority. A progress bar shows the fill rate. Use **Add Supply**.
Facilities	ICP location, base/camp details, staging area manager, and medical aid station info.
Food / Medical (ICS-206)	Meal service log (meal type, count served, menu) plus medical unit leader, ambulance status, hospital contacts.
Check-In (ICS-211)	Personnel check-in list. Records name, ICS position, agency, and arrival time. Links to the Member Roster for pre-populated entries.

20 ICS Finance / Admin Section http://192.168.50.1/ics/finance.html

Finance/Admin manages all financial aspects of the incident: cost accounting, time tracking, procurement, and administrative records for reimbursement.

TAB	HOW TO USE IT
Cost	Log expenditures: category (personnel, equipment, supply, food, transport), amount, description, vendor, and approver. A running total is shown. Use **Add Cost**.
Time	Log person-hours per operator: name, position, agency, hours worked, hourly rate (0 = volunteer). Used for reimbursement claims. Use **Add Time**.
Procurement	Track purchase orders: item, vendor, PO number, amount, status (Requested, Ordered, Delivered, Closed). Use **Add Procurement**.
Admin	Finance/Admin Section Chief info, billing agency, cost-share type, claim deadline, and supporting documentation notes.

■ All finance data can be exported to CSV for submission to the agency finance officer or FEMA reimbursement process. Click **Export CSV** on any Finance tab.

21 NTS Radiogram Generator http://192.168.50.1/nts.html

Generate properly formatted ARRL National Traffic System radiograms. The form produces the standard preamble, address block, and text section, and keeps a traffic log.

Creating a Radiogram

1. Click **Auto-fill Number** for the next sequential message number, or enter one.
2. Set the **Precedence**: EMERGENCY, PRIORITY, WELFARE, or ROUTINE.
3. Choose **Handling Instructions** (HX codes) if needed — HXA through HXF.
4. Enter **Station of Origin**, **Place of Origin**, and click **Auto-fill Date/Time**.
5. Enter the addressee name, address, and phone.
6. Type the message **Text** in ARRL all-caps style. The Check (word count) calculates automatically.
7. Enter the **Signature**, then click **Generate Radiogram**.
8. Click **Save to Log** to keep a copy, then **Print** for the paper record.

Handling Instruction (HX) Codes

CODE	MEANING
HXA	Collect landline delivery authorized up to $ amount
HXB	Cancel message if not delivered within X hours
HXC	Report date and time of delivery to originating station
HXD	Report to originating station the identity of station from which received
HXE	Delivering station get reply from addressee
HXF	Hold delivery until (date)
HXG	Delivery by mail or landline toll call not required

22 ICS-213 General Message http://192.168.50.1/ics213.html

The ICS-213 is the standard written message form for inter- and intra-agency communications on an incident. Use it when you need a written record of a request, situation update, or resource order.

Completing an ICS-213

1. Fill in the header: incident name, To, From, position/agency, date/time, and subject.
2. Click **Auto-fill** to populate the current date and time.
3. Type the message in the body.
4. Click **Generate Form** to render a print-formatted version.
5. Use the **Print** button to produce a field-ready paper copy.
6. Click **Save to Log** to keep a copy in the form log.

> Use the ICS-213 for formal written traffic. For routine spoken radio traffic, use the Net Control log instead.

23 ICS-214 Activity Log & ICS-309 http://192.168.50.1/ics214.html

The ICS-214 is a unit-level activity log required for every ICS section and unit. It records personnel assigned and a timestamped log of activities through the operational period.

Completing an ICS-214

1. Enter the incident name, operational period, unit name, and unit leader.
2. Click **Add Personnel** to list each person assigned to the unit (name, position, agency).
3. For each activity, click **Add Entry**, then **Auto Timestamp** for the current time, and type what happened.
4. Entries accumulate in time order throughout the period.
5. Click **Generate Form** and **Print** for the incident record. **Save Local** keeps a copy on the device.

ICS-309 Communications Log

The companion ICS-309 page (Dashboard → ICS mode → ICS-309 Comms Log) records a formal communications log for the operational period. Each entry has a timestamp, from, to, and message subject. Click **Save to Incident** to file it with the active incident for after-action review.

1. Click **Add (timestamp now)** to add a row with the current UTC time.
2. Fill in From, To, and the message Subject for each entry.
3. Click **Generate Form** to render the printable ICS-309.
4. Click **Save to Incident** to file it in the incident record.

24 Winlink Form Import & Incident Archiving

http://192.168.50.1/winlink-import.html

Winlink Express has its own built-in ICS forms. This chapter explains how to bring that form data into the FieldComms server so the whole incident is documented and archived in one place — and how to re-print a received ICS-213, ICS-214, or ICS-309 on the Pi.

Step-by-Step Import

1. In Winlink Express, open the ICS form message (sent or received).
2. Right-click the **RMS_Express_Form_*.xml** attachment and save it to your computer.
3. On the Winlink Form Import page, drag the saved XML file onto the drop zone, or click Browse.
4. Click **Parse Form Data**. The page detects the form type and extracts all fields.
5. Review the extracted fields in the editable grid. Fix any fields that need correction.
6. Choose the Incident and Direction (Received/Sent), then click ■ **Archive to incident**.

FORM	RE-PRINT ON PI?	NOTES
ICS-213 General Message	Yes	Re-renders as the Pi's ICS-213 for printing
ICS-214 Activity Log	Yes	Activity and personnel lines split into rows automatically
ICS-309 Comms Log	Yes	Log rows parsed into the comms log
Other Winlink forms	Archive only	Data captured and archived — no ICS re-print

Recommended Workflow

1. Run Winlink traffic in Express as your operators normally do.
2. At the end of each operational period, collect any RMS_Express_Form_*.xml files from operators.
3. Import them via the Winlink Form Import page to complete the incident record.
4. Cross-check the ICS-309 log with the Net Control log for completeness.

25 HF Propagation http://192.168.50.1/propagation.html

The propagation page fetches live solar and geomagnetic data from hamqsl.com (when internet is available) and uses it to estimate current HF band conditions.

Reading the Indicators

INDICATOR	WHAT IT MEANS FOR HF
SFI (Solar Flux)	Higher = better high-band propagation. >130 excellent; 90–130 good; <90 40m/80m only.
Sunspot Number	More sunspots = more solar activity = better HF. Follows the 11-year cycle.
A-Index	Daily geomagnetic activity. <10 quiet (good HF); 10–30 unsettled; >30 disturbed.
K-Index (0–9)	3-hour geomagnetic snapshot. ≤2 quiet; 3–4 unsettled; ≥5 storm (HF degraded); ≥7 severe.
X-Ray Flux Class	Solar flare class. A/B/C minor; M moderate (SID possible); X strong HF blackout risk.
Band Condition Bars	Green = good / Amber = fair / Red = poor for each band from 10m through 80m.

> When the Pi has no internet connection, the propagation page shows the last fetched values with a timestamp. For field activations without WAN, check propagation before departing on a device with internet, then switch to the offline Pi for the activation.

26 Repeater Database http://192.168.50.1/repeaters.html

The Repeater Database displays repeater data with rich filtering. It works in three modes: an offline file you load yourself (most reliable), the server API, or built-in sample data. Real repeater data comes from RepeaterBook.com.

The Three Sources

SOURCE	HOW TO USE IT
Offline File (recommended)	Load a RepeaterBook CSV or JSON export. Drag-and-drop the file onto the drop zone, or click Browse. Data persists in your browser after the first load. This needs no API token and is the most dependable method.
Server API	Pulls from the FieldComms server if the net manager has run fetch_repeaters.py to download RepeaterBook data. Requires a RepeaterBook API token.
Sample Data	Three placeholder entries (SAMPLE-1/2/3) for testing the interface. Shows a "not a real repeater" warning. Do not use for operations.

Filtering and Sorting

FILTER	OPTIONS
Band	2m, 70cm, 23cm, 6m, 10m, 1.25m
State	IL, WI, IN, IA (default), or any US state
Affiliation	ARES, SKYWARN, RACES, SATERN, EmComm
Mode	FM, DMR, C4FM/Fusion, D-STAR, P25, NXDN
Sort	Callsign, Frequency, City, Distance (nearest first)

A systemd timer (repeater-refresh.timer) runs the fetch automatically on the first of each month at 04:00, as long as the token is saved in the repeaterbook.env file and the Pi has internet at that time.

27 Facilities Directory http://192.168.50.1/facilities.html

Maintain a directory of EOC locations, hospitals, shelters, staging areas, and command posts. Each entry stores address, coordinates, radio frequencies, CTCSS tone, contact person, on-site callsign, generator status, ADA access, capacity, and operational notes.

Default Facilities

On first startup, FieldComms seeds four McHenry County facilities: the McHenry County EOC (Woodstock), Centegra Hospital Woodstock, the McHenry County Fairgrounds staging area, and Centegra Hospital McHenry. Edit these with your actual operational details.

Managing Facilities

1. Click **+ Add** to create a facility, or click any facility card to view its full detail.
2. In the detail view, click **Edit** to change fields, or **Delete** to remove it.
3. Use **Copy Address** to copy the street address to clipboard for navigation apps.
4. Click **Export CSV** to download the full directory for offline reference.

28 Grid Square Calculator http://192.168.50.1/grid.html

Convert between decimal latitude/longitude and Maidenhead grid squares in both directions, and calculate distance and bearing between two grid squares. Supports 4-character (EN52) and 6-character (EN52ab) precision.

Using the Calculator

1. To convert coordinates to a grid square: enter latitude and longitude, click **Calculate from Lat/Lon**.
2. To convert a grid square to coordinates: enter the grid square, click **Calculate from Grid**.
3. To use your current location: click **Use My Location** (requires location permission, or uses the configured/GPS coordinates).
4. To find distance and bearing: enter two grid squares and click **Calculate Distance**.

The map on the page highlights the grid square on a North America outline for quick visual reference. McHenry County, IL is in grid square **EN52**.

29 Radio Cheat Sheets http://192.168.50.1/cheatsheets.html

A quick-reference set of radio and ICS cheat sheets, organized into tabs. No internet needed — everything is built in.

TAB	CONTENTS
Phonetic Alphabet	NATO phonetics A–Z with pronunciation, plus ITU number pronunciation (Zero, WUN, TOO, TREE, FOW-er, FIFE, SIX, SEV-en, AIT, NIN-er)
Q-Codes	Common amateur Q-codes (QRM, QRN, QRP, QSL, QTH...) and EmComm net Q-codes (QNI, QNS, QND, QNN...)
Prowords	Full procedure word reference — ROGER, WILCO, SAY AGAIN, BREAK, OVER, OUT, CORRECTION, SILENCE, AUTHENTICATE, and more, with meanings and examples
Band Plan	2m and 70cm segments, HF emergency frequencies (ARES, ARRL, IARU, 60m channels), and service bands (MURS, FRS, GMRS, Marine, Aviation)
NTS Precedence	EMERGENCY / PRIORITY / WELFARE / ROUTINE — definitions, when to use each, and examples
ICS Positions	Standard ICS command and general staff titles with abbreviations (IC, OSC, PSC, LSC, FSC, SO, IO, LO)
CTCSS/DCS Tones	Standard CTCSS tone table (67.0–254.1 Hz) and DCS code table for repeater access
Signal Reports	RST and RS signal report codes, S-meter calibration, and signal quality descriptions

30 Print Center http://192.168.50.1/printcenter.html

The Print Center is a one-stop hub for every printable document, plus a built-in incident cover-sheet generator.

CATEGORY	DOCUMENTS
ICS / NTS Forms	ICS-213 General Message, ICS-214 Activity Log, NTS Radiogram, Pre-Flight Checklist
Reference Cards	Phonetic Alphabet, Q-Codes & Prowords, ICS Structure & Forms, CTCSS/DCS & Signal Reports
Operations	Net Control Log (ICS-309), Starcom Net Log (ICS-309), Member Roster, Resource Board
Cover Sheet	Generate a formatted IAP cover sheet from incident details

Generating an Incident Cover Sheet

1. Fill in incident name, number, IC, operational period, frequency, location, and situation summary.
2. Click **Generate Cover** to preview it in the page.
3. Click **Print Cover** to print it in a new window.
4. Click **Clear** to reset the form.

Connecting a Printer to EMCOMM-NET

FieldComms has print buttons on 17 pages. All printing uses the browser standard print function — it sends the job to whatever printer the operator's browser can reach. Three options are supported:

OPTION	HOW IT WORKS	BEST FOR
A — Own printer	Each operator prints to their own locally connected printer. No Pi setup needed.	Single operator with their own printer
B — USB printer via Pi (CUPS)	USB printer plugged into the Pi. CUPS shares it across EMCOMM-NET. Installed automatically. Admin at http://192.168.50.1:631. Supports Windows, Mac, iOS AirPrint, Android, Chromebook.	Field activations — one shared printer for all operators
C — Network printer	Wi-Fi or Ethernet printer connected directly to EMCOMM-NET. Devices discover it via Bonjour/mDNS automatically.	Sites with an existing Wi-Fi capable printer

Setting Up Option B — USB Printer via CUPS

1. Plug the USB printer into the Pi or powered USB hub.
2. Open **http://192.168.50.1:631** from any device on EMCOMM-NET.
3. Click **Administration** → **Add Printer** and log in with the Pi username/password.
4. Select the printer from Local Printers. Check **Share This Printer**. Click Continue.
5. Select the driver, click Add Printer, then print a test page.
6. Windows: Settings → Printers → Add a printer — it appears automatically on EMCOMM-NET.

7. iOS/iPad: tap Share → Print in any FieldComms page — AirPrint finds it automatically.

8. Android: install CUPS Print app → add printer at 192.168.50.1 port 631.

■ Recommended field printers: **Brother HL-L2350DW** (laser, excellent Linux support), **Canon PIXMA TR150** or **HP OfficeJet 200** (battery-powered portable options). Color multifunction laser options for full IAP package printing: **Brother MFC-L3770CDW**, **HP Color LaserJet Pro MFP M479fdw**, or **Canon imageCLASS MF743Cdw** — all support USB, Wi-Fi, and Ethernet. See the Installation Guide Step 8 for full setup instructions and the complete comparison of all three printer connection options.

29.3 Adding the Printer on Each Device

DEVICE TYPE	HOW TO ADD SHARED PRINTER
Windows laptop	Settings → Bluetooth & devices → Printers & scanners → Add a printer. Windows discovers the CUPS printer automatically on EMCOMM-NET via Bonjour.
Mac / macOS	System Settings → Printers & Scanners → click + (Add Printer). Click the Default tab — shared printer appears via Bonjour. Select and click Add.
iPad / iPhone (AirPrint)	No setup required. CUPS + Avahi advertises as AirPrint-compatible. Tap Share → Print in any FieldComms page and select the printer.
Android	Install the free CUPS Print app from Google Play. Add printer at IP 192.168.50.1, Port 631, Protocol IPP.
Raspberry Pi 500 workstation	Chromium discovers CUPS printers via Bonjour automatically. No additional setup needed — printer appears in the print dialog.

31 Reference Library http://192.168.50.1/refs.html

The Reference Library is a document-management system for all field reference materials — radio manuals, interoperability plans, ICS form packages, SOGs, agency plans, and training documents. Files are stored on the Pi and accessible from any device on EMCOMM-NET.

TAB	CONTENTS
■ Amateur Radio	Radio manuals, frequency plans, ARRL publications, band plans, antenna refs
■ ICS / Emergency Mgmt	SWIC plans, NIMS docs, ICS form packages, agency SOGs, mutual aid plans
■ All Documents	Every uploaded document regardless of category

Uploading a Document

1. Click ■ **Upload Document**. A panel slides in from the right.
2. Drag-and-drop a file or click Browse. Accepts PDF, Word, Excel, PowerPoint, images, KML, ZIP, GPX — up to 200 MB.
3. Fill in Title, Category, Source, Description, Tags, Revision, and Expiry (if relevant).
4. Under **Show on tab(s)**, check Amateur Radio, ICS/EmMgmt, or both.
5. Click ■ **Upload Document**. PDFs get an automatic thumbnail.

Finding Documents

1. Use the Category filter, tag cloud, or sort options (Newest, Title, Most Downloaded).
2. Switch between grid and list view using the view toggle.
3. Click any document to open its detail, then Download, Edit, or Delete.

32 Kiwix Offline Library http://192.168.50.1:8081/

Kiwix serves a complete offline reference library from the Pi on port 8081. Every device on EMCOMM-NET can browse it with no internet connection at all. The content is stored as ZIM files — compressed, self-contained web archives that function like offline snapshots of entire websites. WikiMed gives your medical support personnel the same reference content a physician would look up during a mass casualty event. Wikipedia Mini provides general field reference for briefings. iFixit gives your communications team step-by-step repair guides for radio equipment, laptops, and generators — all without internet.

Kiwix has its own built-in full-text search bar that searches across all installed content at once. Type any medical symptom, procedure, equipment model number, or topic and results appear instantly, fully offline. The library is accessible from the dashboard Kiwix Library card, or directly at http://192.168.50.1:8081.

Accessing Kiwix

Open a browser to **http://192.168.50.1:8081**, or tap the Kiwix Library card on the dashboard. Kiwix has a built-in full-text search bar that searches across all installed content at once — type any symptom, procedure, or topic and results appear instantly, offline.

Content Catalogue

TIER	CONTENT	SIZE	BEST FOR
1	WikiMed Medical Encyclopedia	~471 MB	Mass-casualty events, medical support
1	Wikipedia (Mini)	~1.2 GB	General field reference, briefings
1	Wikivoyage	~820 MB	Maps, local facilities, travel info
2	iFixit Repair Guides	~2.3 GB	Equipment repair in the field
2	Wikipedia (Full EN)	~22 GB	Complete reference — large download

Adding a Custom ZIM

1. Download a ZIM from kiwix.org to a device with internet.
2. Copy it to the Pi: `scp mybook.zim fieldcomms@192.168.50.1:/opt/kiwix/zim/`
3. Register it: `sudo kiwix-manage /opt/kiwix/library.xml add /opt/kiwix/zim/mybook.zim`
4. Restart: `sudo systemctl restart kiwix`
5. Open http://192.168.50.1:8081 — the new content appears.

33 Offline Maps, GPS & Health Monitor

This chapter covers three infrastructure features that operate in the background to support the rest of FieldComms: the offline map tile system that powers both the Tactical APRS Map and the Starcom Resource Map, the GPS live position feed that provides accurate coordinates across all tools, and the Health Monitor that tracks system vitals and service status in real time.

32.1 Offline Map Tile System

Map tiles are downloaded in advance, stored as MBTiles files on the Pi, and served locally on port 8083. Both the Tactical APRS Map and Starcom Resource Map use offline tiles by default. The default source is USGS Imagery+Topo Hybrid — public-domain satellite imagery with roads, contours, and place names.

Downloading Tiles

1. Run `sudo bash /opt/fieldcomms/scripts/download_tiles.sh` for the interactive menu (106 presets: all Illinois counties, WI/IN/IA border counties, all 50 states).
2. Search presets: `download_tiles.sh --search "mchenry"`
3. Download a preset: `download_tiles.sh --area "McHenry County IL" --zoom 8-16`
4. Tiles are stored in /opt/fieldcomms/data/tiles/ and served by the tile server on port 8083.

32.2 GPS Live Position

If a USB GPS receiver is connected to the Pi (via the powered USB hub), gpsd provides live coordinates to the tactical map, health monitor, and NWS alerts. The GPS receiver creates a stable **/dev/gps0** device via udev rules installed automatically.

CHECK	COMMAND
GPS status	gpspipe -w -n 5
GPS device	ls -la /dev/gps0
gpsd status	sudo systemctl status gpsd
Live fix data	cgps -s

32.3 Health Monitor

The health monitor runs as a background service on port 5051 and feeds the dashboard sidebar with live system diagnostics. Raw data is at **http://192.168.50.1:5051/health**.

METRIC	NORMAL	ACTION IF EXCEEDED
CPU Temperature	< 70°C	Improve ventilation. Pi 5 throttles at 85°C.
Memory Usage	< 80%	Close unused browser tabs; restart non-essential services
Disk Usage	< 90%	Archive old logs; remove unused ZIM files

METRIC	NORMAL	ACTION IF EXCEEDED
Service Status	All green	Red dot = service down. Restart with systemctl.
Internet	Connected (if avail.)	Check Ethernet. Core features work offline.
GPS	Fix (if attached)	No GPS is fine — coordinates configured at install.

Restarting a Stopped Service

```
sudo systemctl restart fcc-lookup # or: health-monitor, ics-platform, fieldcomms-refs, kiwix, deadmans
```

34 Network Hardware — ASUS RT-BE58 Go & UniFi Switch Lite 16

FieldComms v1.0 uses a three-router mesh network rather than a single access point. The primary ASUS RT-BE58 Go travel router manages WAN connectivity, DHCP, and the EMCOMM-NET Wi-Fi access point. Two additional RT-BE58 Go routers operate as AiMesh nodes extending EMCOMM-NET to secondary rooms, outdoor staging areas, and upper floors — all sharing the same SSID and password with seamless device roaming. The Pi itself never broadcasts Wi-Fi. It communicates over wired Ethernet only.

DEVICE	ROLE	CONNECTION
ASUS RT-BE58 Go (primary)	Wi-Fi 7 AP · DHCP server · WAN gateway · AiMesh controller	WAN: InstyConnect cellular · USB WAN: Starlink failover · LAN: UniFi switch
ASUS RT-BE58 Go (mesh node 1)	EMCOMM-NET extension — same SSID, seamless roaming	Wired backhaul via UniFi Switch Port 11
ASUS RT-BE58 Go (mesh node 2)	EMCOMM-NET extension — third coverage zone	Wired backhaul via UniFi Switch Port 12
UniFi Switch Lite 16 PoE	16-port GbE managed switch · 8x PoE · wired distribution hub	Port 1: router uplink · Ports 2-12: devices · Ports 13-16: spare

33.1 EMCOMM-NET Wi-Fi Coverage

All three routers broadcast EMCOMM-NET on 2.4 GHz and 5 GHz simultaneously. Devices connect to whichever band and router provides the strongest signal and roam automatically as operators move around the venue. A single RT-BE58 Go covers approximately 2,000 to 2,500 square feet indoors. With both mesh nodes active, the system covers 7,500 to 20,000 square feet depending on building construction. For outdoor SAR staging areas, a node on battery at the perimeter extends EMCOMM-NET to field positions without additional cabling.

33.2 WAN Connectivity — InstyConnect Primary + Starlink Failover

The ASUS router manages two WAN sources with automatic failover. InstyConnect cellular (T-Mobile and Verizon dual-carrier) is always the primary. Starlink satellite is secondary — the ASUS switches to it automatically within 30 to 60 seconds when cellular drops, and switches back when cellular recovers. No operator action is required for the failover in either direction.

PRIORITY	WAN SOURCE	HOW CONNECTED	ACTIVATES
1 — Primary	InstyConnect cellular (T-Mobile + Verizon)	PoE Ethernet from Drum or Switchblade antenna to ASUS WAN port	Always — default for all activations
2 — Auto failover	Starlink satellite	Starlink Ethernet adapter → USB-Ethernet adapter → ASUS USB WAN port	Automatic when cellular WAN drops or degrades
3 — Manual	EOC site Ethernet	Site cable to ASUS WAN port (replaces InstyConnect cable)	Manual — when reliable site internet is available

PRIORITY	WAN SOURCE	HOW CONNECTED	ACTIVATES
4 — Last resort	USB smartphone tether or venue Wi-Fi (WISP)	Phone USB to ASUS USB port or ASUS WISP wireless WAN mode	Manual — emergency fallback only

33.3 InstyConnect Antennas

FieldComms carries two InstyConnect antennas to handle different deployment sites. Both connect via a single outdoor-rated PoE Ethernet cable to the ASUS WAN port. The modem is integrated into the outdoor antenna enclosure — no separate modem box is needed. Power travels up the same cable that carries data.

ANTENNA	TYPE	WHEN TO USE	AIMING
InstyConnect Drum	Omnidirectional 5G/LTE	Default antenna — deploy at every activation. Mounts on any 1.5" to 2" mast, tripod, or vehicle roof rack.	No aiming needed — omnidirectional picks up all towers from all directions.
InstyConnect Switchblade	Directional 4x folding LDAP	When Drum signal is insufficient at a specific site. Folds flat for transport. Deploys in under 5 minutes. Swap the PoE cable from the Drum to the Switchblade on the ASUS WAN port.	Aim toward nearest tower using the InstyConnect signal app. Rotate slowly while watching signal bars — stop at peak signal.

InstyConnect data plan management: Keep the plan in Standby mode ($5/month) between activations. Standby maintains the SIM and multi-network capability so you can activate at full speed the moment an emergency is declared. Activate and pause plans at instyconnect.com. Active plan cost is approximately $79-99/month on the Multi-Network Unlimited plan.

33.4 WAN Status Dashboard

The WAN Status Dashboard at http://192.168.50.1/wan-status.html gives operators a live view of all WAN sources from any browser on EMCOMM-NET. The active WAN source is shown in a large color-coded panel: green for InstyConnect cellular, blue for Starlink satellite, red for offline. InstyConnect signal strength, carrier, and technology are displayed. Starlink latency, throughput, and obstruction percentage appear when the dish is active. A connectivity test panel verifies reachability to NWS weather alerts, APRS-IS, Cloudflare DNS, and the AMPRNet gateway every 60 seconds. A WAN event log records every failover and failback with UTC timestamps.

33.5 AMPRNet / 44Net Gateway

A dedicated Raspberry Pi 5 at 192.168.50.2 runs the AMPRNet gateway service. It maintains a permanent WireGuard encrypted tunnel to amprgw.ampr.org and routes the entire 44.0.0.0/8 AMPRNet address block for all EMCOMM-NET devices. This means any device on EMCOMM-NET — phones, tablets, operator workstations, and the FieldComms Pi itself — can reach other AMPRNet stations globally without routing traffic through the commercial internet.

AMPRNET USE CASE	HOW IT WORKS FROM EMCOMM-NET
Winlink via AMPRNet	Pat Winlink on the FieldComms Pi can connect to RMS gateways on 44.x.x.x addresses directly over AMPRNet — bypassing commercial internet when only amateur radio paths are available.

AMPRNET USE CASE	HOW IT WORKS FROM EMCOMM-NET
APRS-IS via AMPRNet	Graywolf and YAAC can connect to APRS-IS servers on 44.x.x.x — keeping APRS traffic within the amateur radio network.
Inter-node FieldComms	If a second MCESV FieldComms system is deployed with its own 44Net gateway, the two systems can share data over AMPRNet without commercial internet routing.
Direct amateur station connectivity	Any licensed amateur station worldwide with a 44.x.x.x address is directly reachable from any EMCOMM-NET device for data exchange, file transfer, or status checking.

The gateway status page at http://192.168.50.2:9000 shows tunnel state, last handshake time, AMPRNet address, bytes transferred, and system health of the gateway Pi. Access requires a valid FCC amateur radio callsign. The FieldComms dashboard health monitor also shows a 44Net status indicator updated every 30 seconds by the amprgate-poll background service.

33.6 AMPRNet Gateway — Access Control

Access to the AMPRNet gateway is restricted to FCC-licensed amateur radio operators. This is both a security measure and a Part 97 compliance requirement. The gateway uses a two-level access model:

LEVEL	HOW TO ACCESS	WHAT YOU CAN DO
Status dashboard (any EMCOMM-NET device)	Open http://192.168.50.2:9000 in any browser on EMCOMM-NET. Enter your FCC callsign when prompted. Callsign is validated against the local FCC database on the FieldComms Pi.	View tunnel state, AMPRNet address, last handshake, traffic statistics, routes, and gateway system health. Read-only — no tunnel control from this level.
Tunnel control (gateway Pi keyboard only)	Sit at the gateway Pi keyboard. Open Chromium to http://localhost:9001. Log in with your FCC callsign. Physical presence at the gateway Pi is required — this port is blocked from the network.	Bring the WireGuard tunnel up, down, or restart it. All actions are logged with callsign, timestamp, and IP address.

> The /api/status endpoint on port 9000 does not require authentication. This is intentional — the FieldComms Pi poll service reads it every 30 seconds as a machine-to-machine call to update the dashboard WAN indicator. This endpoint is read-only and returns no sensitive data. All web UI access and all tunnel control actions are logged to /var/log/amprgate-access.log on the gateway Pi.

33.7 Network IP Reference

DEVICE	IP ADDRESS	ADMIN URL
FieldComms Pi 5 (application server)	192.168.50.1	http://192.168.50.1
44Net Gateway Pi 5	192.168.50.2	http://192.168.50.2:9000
ASUS RT-BE58 Go (primary router)	192.168.50.254	http://192.168.50.254
Windows Laptop (recommended)	192.168.50.3	DHCP reservation in router
Color MFP Printer (recommended)	192.168.50.10	DHCP reservation in router

DEVICE	IP ADDRESS	ADMIN URL
Pi 500 Workstations (×4)	192.168.50.20–23	DHCP reservations in router
InstyConnect modem	10.1.1.1	http://10.1.1.1 (via InstyConnect Wi-Fi)
Starlink dish admin	192.168.100.1	http://192.168.100.1 (via Starlink network)
Field devices (DHCP)	192.168.50.100–200	Assigned automatically by ASUS router
CUPS print server	192.168.50.1	http://192.168.50.1:631
Kiwix offline library	192.168.50.1	http://192.168.50.1:8081
Pat Winlink	192.168.50.1	http://192.168.50.1:8090
Health Monitor API	192.168.50.1	http://192.168.50.1:5050/health
WAN Status Dashboard	192.168.50.1	http://192.168.50.1/wan-status.html
AMPRNet Gateway Status	192.168.50.2	http://192.168.50.2:9000

35 JS8Call — HF Digital Keyboard Messaging (Windows)

http://192.168.50.1/js8call

JS8Call is a keyboard-to-keyboard HF digital messaging mode based on FT8 technology. It is designed for very weak signal conditions — stations can exchange readable messages at signal levels 20 dB below what voice would require. JS8Call supports direct messaging between two stations, group messaging to a named group callsign such as @EMCOMM, store-and-forward relay through intermediate stations when no direct path exists, and heartbeat beacons that advertise a station presence automatically.

JS8Call runs on the Windows laptop alongside Winlink Express, both connected to the IC-7300 via the single USB cable. The FieldComms dashboard provides a purple JS8Call quick-launch card that opens JS8Call built-in web interface from any device on EMCOMM-NET — operators at the EOC table can monitor JS8Call traffic from their phones or tablets without being at the Windows laptop.

34.1 Recommended JS8Call Frequencies

BAND	DIAL FREQUENCY	NOTES
40m (primary for ARES/RACES)	7.078.0 MHz	Primary JS8Call calling frequency — most activity
80m (regional, nighttime)	3.578.0 MHz	Good regional coverage after dark
20m (long distance)	14.078.0 MHz	DX and national nets — good during daylight
17m (secondary)	18.104.0 MHz	Less congested alternative to 20m
60m (emergency designated)	5.357.0 MHz USB	FCC Part 97.303(h) emergency channels

34.2 JS8Call for EmComm

USE CASE	HOW TO DO IT
Direct station message	Type @[CALLSIGN]: followed by your message. The addressed station receives it even if they are not actively watching — JS8Call stores incoming messages.
Group message (all EMCOMM stations)	Type @EMCOMM: followed by your message. All stations monitoring the @EMCOMM group receive it and it appears highlighted in their message window.
Heartbeat beacon (announce presence)	Enable Heartbeat in JS8Call settings. Your station transmits a short beacon every 15 minutes advertising your callsign, grid, and signal strength.
Store-and-forward relay	If direct contact is not possible, any intermediate JS8Call station on the path can relay your message to the destination automatically.
Query station SNR	Send CALLSIGN SNR? to ask a station to report your signal strength. Useful for propagation assessment before attempting Winlink.

36 ICS Planning P — Operational Planning Cycle

http://192.168.50.1/ics/planningp.html

The Planning P page is an interactive guide to the ICS operational planning cycle. It presents all 15 phases of the Planning P — from the initial incident through each operational period — with the standard agenda, required ICS forms, and attendee list for every phase. Access it from the ■ **Planning P** tab on any ICS platform page.

The 15 Phases

PHASE	NAME	KEY FORMS
1	Incident / Event	ICS-201 (begin)
2	Notification	ICS-201
3	Initial Response & Assessment	ICS-201
4	Initial UC Meeting	ICS-201, ICS-202
5	Incident Brief — ICS-201	ICS-201
6	IC/UC Develop/Update Objectives Meeting	ICS-202
7	Command & General Staff Meeting/Briefing	ICS-202, ICS-203
8	Preparing for the Tactics Meeting	ICS-215, ICS-215A
9	Tactics Meeting	ICS-215, ICS-215A, ICS-203, ICS-204
10	Preparing for the Planning Meeting	ICS-204, ICS-205, ICS-206, ICS-207, ICS-208
11	Planning Meeting	ICS-202 through ICS-206
12	IAP Prep & Approval	ICS-202 through ICS-208 (complete IAP)
13	Operations Briefing	Complete IAP, ICS-204 by Division
14	Execute Plan & Assess Progress	ICS-214, ICS-209, ICS-309
15	New Ops Period Begins	ICS-214, ICS-209

Using the Planning P Page

1. Open the ICS Platform (http://192.168.50.1/ics/) and click the ■ **Planning P** tab.
2. The reference image of the official Planning P diagram is shown on the left for reference.
3. The 15 phase buttons are listed in the center panel, organized into five color-coded groups.
4. Click any phase button to see its details in the right panel.

Phase Detail Panel

SECTION	WHAT IT SHOWS
Standard Agenda	Numbered list of the standard agenda items for that meeting or activity
Required Forms	ICS form numbers as clickable chips — click to open the form directly
Who Should Attend	List of ICS roles required or optional for that phase, with red/green roster dots showing who is currently checked in

The Phase Color Groups

COLOR	GROUP	PHASES	ICS PHASE OF PLANNING
Gray	Initial Response	1–5	Stem — initial incident response
Yellow	Establish Objectives	6–7	Stem — incident objectives set by IC/UC
Red	Develop the Plan	8–11	Loop — tactics through planning meeting
Green	Prepare & Disseminate	12–13	Loop — IAP approval and ops briefing
Teal	Execute, Evaluate & Revise	14–15	Loop — field operations and next period

Generating a Briefing Sheet

1. Select any phase by clicking its button.
2. Click **Generate briefing sheet** in the detail panel.
3. A printable one-page cover sheet opens with the phase name, agenda, required forms, and an attendance table with signature lines.
4. Print it for the IAP package or the planning meeting folder.

> The Planning P page is a guide, not a workflow enforcer. You do not need to click through the phases in order. Use it as a quick reference during planning meetings to ensure nothing is missed — especially on activations where the planning cycle is compressed.

34 Appendix — Administration & Quick Reference

A1 Installation & Updates

COMMAND	WHAT IT DOES
sudo bash install.sh	Interactive installer — sets callsign, coordinates, Wi-Fi, services
sudo bash update.sh	Updates FieldComms code and restarts services
sudo bash kiwix_setup.sh	Install/manage Kiwix offline content
sudo bash download_tiles.sh	Download offline map tiles

A2 Wi-Fi & Network

ITEM	DETAIL
SSID	EMCOMM-NET
Pi IP	192.168.50.1 (static, set via nmcli on eth0)
DHCP range	192.168.50.100 – 192.168.50.200
Router admin	http://192.168.50.1 (ASUS RT-BE58 Go)
CUPS admin	http://192.168.50.1:631 (printer management)

A3 Database Backup

The main database is a single SQLite file. Back it up by copying **fieldcomms.db** to a USB drive. Export the roster CSV monthly and keep a copy on a USB drive labeled FIELDCOMMS as a secondary backup. Plugging in a drive labeled FIELDCOMMS automatically triggers a full rsync backup of the application data via the udev backup rule.

A4 Winlink — Email Over Radio

MCESV/MCEMA uses **Winlink Express** as the primary Winlink client on the Windows laptop with the IC-7300 + VARA HF. The Pi also runs **Pat** as a backup browser-based Winlink client on port 8090. Never run both clients with an active session on the same radio simultaneously.

A5 Service & Port Reference

SERVICE	PORT	DESCRIPTION
nginx	80	Web server — serves all HTML pages
fcc-lookup.service	5050	FCC callsign API, net log, forms, DMS, roster, resources
health-monitor.service	5051	System health API

SERVICE	PORT	DESCRIPTION
ics-platform.service	5055	ICS incident platform API
fieldcomms-refs.service	5056	Reference library API
Graywolf APRS	8080	APRS client with REST API and WebSocket
Kiwix	8081	Offline library — WikiMed, Wikipedia, iFixit
YAAC APRS	8082	Secondary APRS client
Tile server	8083	Offline map tile server (MBTiles)
Pat Winlink	8090	Browser-based backup Winlink client
CUPS printer	631	Print server — USB printer shared to EMCOMM-NET
JS8Call (Windows)	2442	JS8Call TCP API on Windows laptop

Operator Workstation Options — Raspberry Pi 500 / 500+ + Raspberry Pi Monitor

FieldComms is a browser-based system — any device with a modern browser and Wi-Fi can connect to EMCOMM-NET and access every tool. The Raspberry Pi 500 paired with the official Raspberry Pi Monitor provides a purpose-built, low-cost, low-power operator workstation that is fully Pi ecosystem compatible and ideal for fixed EOC positions, shelter check-in stations, and net control operator desks. Funding through ARPA-E grants, FEMA BRIC, or 501(c)(3) equipment donations may cover these workstations.

Pi 500 vs Pi 500+ vs Pi 400 — Comparison

SPEC	Pi 400 (2020)	Pi 500 (Dec 2024) ★ Recommended	Pi 500+ (Sept 2025)
Processor	Cortex-A72 @ 1.8 GHz (Pi 4 generation)	**Cortex-A76 @ 2.4 GHz (Pi 5 generation)**	Cortex-A76 @ 2.4 GHz (Pi 5 generation)
RAM	4 GB DDR4	**8 GB LPDDR4X**	16 GB LPDDR4X
Storage	microSD only	microSD only (32 GB A2 included)	256 GB NVMe built-in + microSD slot
Keyboard	Membrane	Membrane	Mechanical (Gateron Blue KS-33)
Wi-Fi	802.11ac dual-band	802.11ac dual-band	802.11ac dual-band
Ethernet	Gigabit	Gigabit	Gigabit
USB ports	2× USB 3.0, 1× USB 2.0	2× USB 3.0, 1× USB 2.0	2× USB 3.0, 1× USB 2.0
Display output	2× micro-HDMI	2× micro-HDMI	2× micro-HDMI
Production until	Jan 2028	**TBD (current model)**	TBD (current model)
Best for FieldComms?	Acceptable — older chip, only 4 GB RAM	**✓ YES — Pi 5 chip, 8 GB RAM, current generation**	Overkill for browser use Justified only if also running heavy Linux apps locally

Raspberry Pi Monitor — 15.6"

SPEC	DETAIL
Panel	15.6" IPS LCD, 1920×1080 Full HD @ 60 Hz
Video input	Standard HDMI 1.4 (full-size) — requires micro-HDMI to HDMI cable with Pi 500
Audio	2× front-facing stereo speakers, 3.5mm headphone jack
Power	USB-C, 5V/1.5A — can power from Pi 500 USB port at 60% brightness/volume, or from separate USB-C supply for 100% brightness/volume
Mounting	Integrated angle-adjustable kickstand, VESA mount, wall hanger
Dimensions	Slim 15.6" form — similar footprint to a laptop
Colors	Red/White (matching Pi 500 keyboard) or Black
Where to buy	raspberrypi.com · Adafruit · PiShop.us · GigaParts · Amazon

■ The Pi 500 has micro-HDMI ports; the Raspberry Pi Monitor has a standard full-size HDMI input. You need a micro-HDMI to standard HDMI cable (or adapter). The Pi 500 desktop kit ($120) includes one. When ordering separately, confirm the cable is included or purchase one — they are available from the same retailers listed above.

Operator Workstation — Bill of Materials (Per Station)

ITEM	MODEL / SPEC	WHERE TO BUY
— Standard Workstation (Pi 500) —		
Raspberry Pi 500 keyboard computer	Pi 500 standalone — Pi 5 chip, 8 GB RAM, 32 GB A2 microSD, keyboard	raspberrypi.com Adafruit · Amazon
Raspberry Pi Monitor	15.6" Full HD IPS, built-in speakers, kickstand, VESA mount	raspberrypi.com GigaParts · Amazon
micro-HDMI to HDMI cable	1m micro-HDMI to standard HDMI (Pi 500 → Monitor) Included in Pi 500 Kit ($120) — verify before ordering separately	raspberrypi.com Amazon
USB mouse	Any USB or Bluetooth mouse — included in Pi 500 Kit	Amazon
USB-C power supply for Pi 500	Official Raspberry Pi 27W USB-C PD power supply (or equivalent)	raspberrypi.com Amazon
USB-C power supply for monitor (optional, for full brightness)	5V/1.5A USB-C supply — skip if powering monitor from Pi USB port (Pi USB port gives 60% brightness which may be dim outdoors)	Amazon
— Premium Workstation (Pi 500+) —		
Raspberry Pi 500+ keyboard computer	Pi 500+ — Pi 5 chip, 16 GB RAM, 256 GB NVMe SSD, mechanical keyboard (Gateron Blue), RGB	raspberrypi.com Adafruit · Amazon
Raspberry Pi Monitor + cables + mouse	Same as Standard Workstation above	Same as above

■ **501(c)(3) Grant Funding Note:** MCESV/MCEMA is a 501(c)(3) nonprofit. Operator workstation equipment may be eligible for funding through: FEMA BRIC (Building Resilient Infrastructure & Communities), ARPA-E Emergency Communications grants, ARRL Foundation grants, local Community Foundation equipment grants, or AUXCOMM/ARES regional funding. Document each workstation as "emergency communications operator terminal for the McHenry County Emergency Services Volunteers (K9ESV), a 501(c)(3) nonprofit supporting McHenry County Emergency Management Agency (MCEMA) RACES/ARES/Starcom communications."

Recommended Deployment by Position

POSITION	RECOMMENDED DEVICE	REASON
Net Control Operator (fixed EOC desk)	**Pi 500 + Pi Monitor**	Dedicated keyboard, good display, Chromium runs every FieldComms page smoothly
ICS Section Chief (Planning, Ops, Logistics)	Pi 500 + Pi Monitor or Windows/Mac laptop	Needs reliable keyboard for ICS form entry; Pi 500 is sufficient for all ICS platform pages
Served Agency Staff (EOC or shelter)	Any tablet, Chromebook, or Pi 500	Observer Mode and ICS forms work on any browser — no special hardware needed
Winlink / JS8Call Operator	Windows laptop (required)	Winlink Express and JS8Call are Windows applications — Pi 500 cannot run them natively

Mobile / Field Operator	Phone or tablet	FieldComms works on any smartphone browser on EMCOMM-NET — no dedicated hardware needed

■ The Pi 500 cannot run Windows applications natively. Winlink Express and JS8Call must run on the Windows laptop connected to the IC-7300. The Pi 500 is ideal for all browser-based FieldComms tools — net control logging, ICS forms, tactical maps, resource tracking, and the full ICS platform.